

E-Commerce Business Analysis

SQL + PostgreSQL + Power BI Analytics Project

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Dataset **Brazilian Olist E-Commerce**

Sales · Customers · Delivery · Payments · Sellers

Project Objective

01

Analyse E-Commerce Performance

Deep-dive across sales, customers, delivery, and payments on 99,441 real orders to uncover meaningful business patterns.

02

Transform Raw Data into Insights

Converted 6 raw transactional tables into clean, business-ready intelligence using PostgreSQL and 33 analytical queries.

03

Build Reusable SQL Views

Engineered 6 modular SQL views (vw_monthly_orders, vw_state_revenue etc.) feeding directly into Power BI dashboards.

04

Present Business Recommendations

Translated findings into 5 actionable recommendations with quantified impact for e-commerce decision-makers.

Data & Workflow

Raw Data
(6 tables)



PostgreSQL
Analysis
(33 queries)



SQL Views
(6 views)



1 Power BI
Dashboard

Tables Used

orders

customers

order_items

payments

products

sellers

SQL Views Created

▸ vw_monthly_orders

▸ vw_delivery_performance

▸ vw_state_revenue

▸ vw_seller_product_perf

▸ vw_customer_insights

▸ vw_payment_analysis

Year

2016

2017

2018

Ecommerce Business analysis

99K

total_orders

13M

total_revenue

93K

Unique customers

12

delivery_days

customer_state

AC

AL

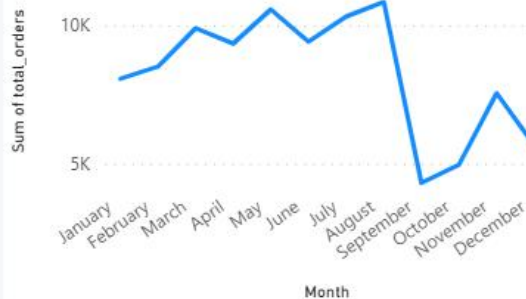
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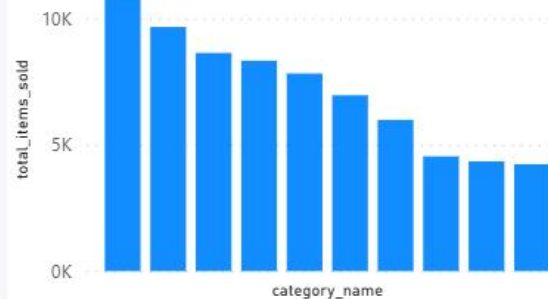
Monthly order count



Area Distribution



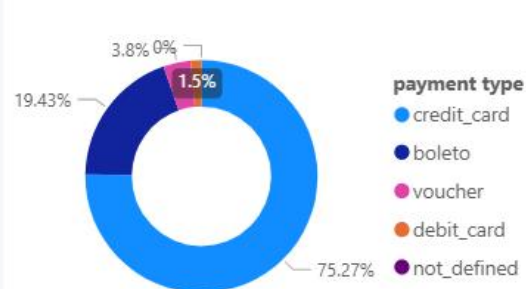
Top Performing Product Categories



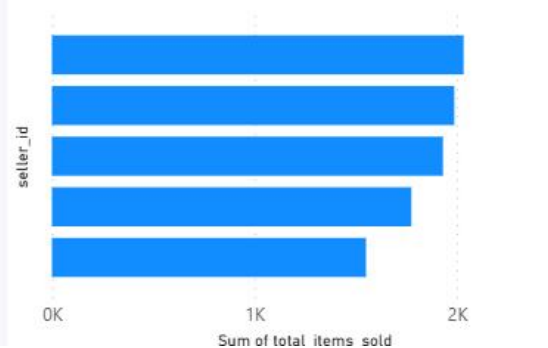
Delivery Analysis



Payment types distribution



Top 5 sellers by sales



SQL Skills Demonstrated

Joins & Aggregations

Multi-table joins across 6 tables — orders, customers, order_items, sellers, products, payments — with grouped aggregations.

```
JOIN customers c ON o.customer_id =  
c.customer_id
```

CTEs & CASE WHEN

Common Table Expressions for modular logic; CASE WHEN for customer segmentation, delivery speed buckets, seller tiers.

```
WITH monthly_revenue AS (...) SELECT  
month, total_revenue, LAG...
```

Window Functions

ROW_NUMBER() OVER PARTITION for top-3 customers per state; LAG() for month-on-month revenue growth calculation.

```
ROW_NUMBER() OVER (PARTITION BY states  
ORDER BY spending DESC)
```

Date & Timestamp

EXTRACT(EPOCH) for delivery time in days; DATE_TRUNC for monthly aggregation; NULLIF for empty string handling.

```
avg(nullif(delivered_date, '')::timestamp  
- purchase_timestamp)
```

NULL Handling

NULLIF + IS NOT NULL chains to robustly handle empty strings in delivery date fields across 99,441 rows.

```
WHERE  
NULLIF(order_delivered_customer_date, '')  
IS NOT NULL
```

View Engineering

6 reusable PostgreSQL views created as dashboard-ready data layers — separating business logic from reporting layer.

```
CREATE VIEW vw_state_revenue AS SELECT  
c.customer_state, SUM...
```

Sales & Revenue Insights

99,441

Total Orders

R\$ 13.6M

Total Revenue

96,478

Delivered (97%)

625

Cancelled Orders

Revenue Concentration by State

State	Orders	Revenue	% of Total
SP	41,746	R\$ 5.2M	38.2%
RJ	12,852	R\$ 1.8M	13.2%
MG	11,635	R\$ 1.6M	11.7%
RS	5,466	R\$ 0.75M	5.5%
PR	5,045	R\$ 0.68M	5.0%

SP + RJ + MG = 63.1% of all revenue

Month-on-Month Revenue Growth

Jan 2017	R\$ 120K	+1,103,688%*
Feb 2017	R\$ 247K	+105.6%
Mar 2017	R\$ 374K	+51.4%
Apr 2017	R\$ 360K	-3.9%
May 2017	R\$ 506K	+40.6%

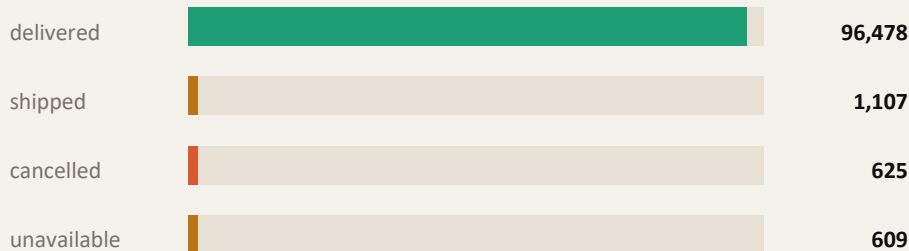
*Jan 2017 spike due to near-zero Dec 2016 base — platform was in early launch phase

Customer & Delivery Insights

Customer Behaviour

- 99,441 unique customers — 1:1 ratio with orders. Zero repeat customers detected in this dataset.
- Top customer spent R\$ 13,440 across multiple high-value orders — 2x the next highest spender.
- Top 3 customers by spend: R\$ 13,440 · R\$ 7,160 · R\$ 6,735 — indicating high-value individual buyers.
- Business interpretation: customer retention is a critical gap. Lifetime value strategy needed.

Order Status Breakdown



Delivery Performance

Avg delivery time

12.5 days

Delivered orders

96,478 (97%)

Cancelled orders

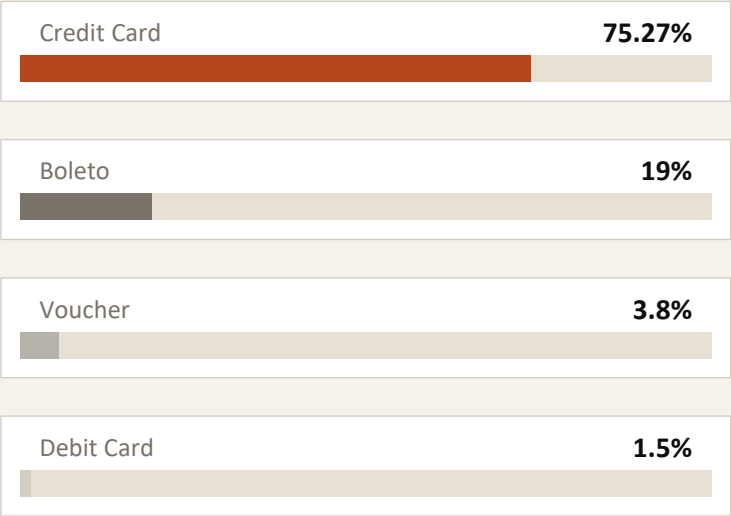
625 (0.6%)

Avg time before cancellation

19.5 days

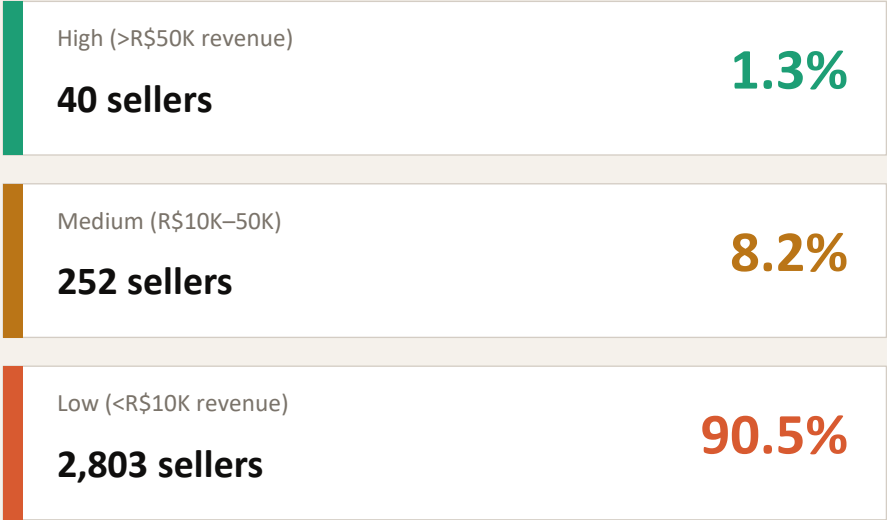
Payment & Seller Insights

Payment Distribution



Credit card dominates at 73.9%. Boleto (offline bank payment) at 19% signals significant non-digital user base.

Seller Performance Segmentation



Top seller: 2,033 orders. 3,096 total sellers. Top 40 high-performers drive majority of GMV.

Top Product Categories

cama_mesa_ban (Bed & Bath) #1 volume #3 revenue	beleza_saude (Health & Beauty) #2 volume #1 revenue	relogios_present (Watches & Gifts) #7 volume #2 revenue
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Business Recommendations

01

Double down on SP, RJ, MG

63.1% of revenue from 3 states. Direct marketing, faster delivery SLAs, and seller recruitment investment in these zones first before expanding.

02

Urgent: build customer retention

Zero repeat customers detected. Implement loyalty programme, post-delivery follow-up, and personalised re-engagement — average top customer spent R\$ 13,440.

03

Reduce delivery delays in tail regions

Average delivered = 12.5 days. Cancelled orders took 19.5 days before cancellation — customers waited too long. Target sub-10 day delivery in top 5 states.

04

Elevate top 40 high-performing sellers

Only 40 of 3,096 sellers earn >R\$50K. Invest in seller success programmes — premium onboarding, dedicated support, and marketing co-investment for the top tier.

05

Promote Boleto & instalment payments

19% Boleto usage signals offline-first users. Offering EMI/instalment options can increase basket size, especially for watches and electronics categories.

Key Findings at a Glance

Summary of what the data revealed

63.1%

of revenue from SP, RJ & MG alone

SP = R\$5.2M · RJ = R\$1.8M · MG = R\$1.6M

0

repeat customers detected

Every single order came from a unique customer — major retention gap

12.5

average days to deliver

97% of 99,441 orders successfully delivered. Cancelled orders waited 19.5 days.

73.9%

payments via credit card

Boleto at 19% — significant non-digital user base requiring offline payment support

40/3096

sellers classified as high performers

Only 1.3% of sellers earn >R\$50K. Top seller: 2,033 orders. Huge seller quality gap.

R\$13,440

highest single customer spend

Top 3 customers: R\$13,440 · R\$7,160 · R\$6,735 — strong high-value segment

Thank You

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Dataset: Brazilian Olist E-Commerce (Kaggle) · Tools: PostgreSQL, Power BI